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ENCLOSURE FOR ISOLATED CONVERTER AND ISOLATED CONVERTER USING SAME

Abstract

An enclosure for an isolated converter and an isolated converter using the same are provided. The enclosure for an isolated converter, according to an embodiment of the present invention, may include a main body having a primary-side space and a secondary-side space. The main body may include a partition wall provided to vertically separate the primary-side space and the secondary-side space from each other, and an upper wall and a lower wall provided above and below the partition wall, respectively.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATION
[0001] This application is a National Stage of International Application No. PCT/KR2022/002154, filed on Feb. 14, 2022, which claims priority to and the benefit of Korean Patent Application No. 10-2021-0038987, filed on Mar. 25, 2021, the disclosure of which is incorporated herein by reference in its entirety.
FIELD
[0002] The present disclosure relates to an enclosure for an isolated converter and an isolated converter using the same.
BACKGROUND
[0003] In general, a DC-DC converter or an AC-DC converter is provided in a switching mode power supply (SMPS) to stably supply control power in a high voltage environment. A converter (DC-DC or AC-DC converter) is a device that arbitrarily varies the output voltage with respect to the input power. Here, the converter has a primary-side input and a secondary-side output.
[0004] In addition, the converter can be divided into an isolated type that provides an isolation function between input/output voltages and a non-isolated type that does not provide this. Here, the isolation performance of the isolated converter is determined according to the structure of the enclosure.
[0005] In an isolated SMPS or an isolated converter, ‘isolated’ means to include a transformer structure consisting of primary and secondary coils that are magnetically connected but electrically separated/isolated. Therefore, it is important to maintain electrical isolation between the primary coil and the secondary coil at the time of high voltage input.
[0006] Meanwhile, the isolated converter is required to reduce manufacturing costs and miniaturize for installation and operation. However, the conventional isolated converter has a problem in that it cannot simultaneously satisfy both isolation performance and minimization of the overall size due to the structure of the enclosure.
SUMMARY
[0007] The present disclosure is directed to providing an enclosure for an isolated converter and an isolated converter using the same, minimizing the overall size while maintaining isolation performance.
[0008] In order to solve the above technical problems, the enclosure for an isolated converter according to an embodiment of the present disclosure may include a main body having a primary-side space and a secondary-side space. Here, the main body may include a partition wall provided to vertically separate the primary-side space and the secondary-side space from each other; and an upper wall and a lower wall provided above and below the partition wall, respectively.
[0009] In an embodiment of the present disclosure, the partition wall may be provided in the center of the main body.
[0010] The enclosure for an isolated converter according to an embodiment of the present disclosure may further include side plates provided on both sides of the main body.
[0011] The enclosure for an isolated converter according to an embodiment of the present disclosure may further include input/output terminals provided on the outside of the side plates.
[0012] In an embodiment of the present disclosure, the primary-side space and the secondary-side space may be defined by the upper wall, the lower wall and the partition wall.
[0013] The enclosure for an isolated converter according to an embodiment of the present disclosure may further include a first stepped portion provided from the partition wall to form a step difference with the upper wall and the lower wall, in each of the primary-side space and the secondary-side space.

[0014] The enclosure for an isolated converter according to an embodiment of the present disclosure may further include a second stepped portion provided from the first stepped portion to the opposite side of the partition wall and having a groove portion in the center, in each of the primary-side space and the secondary-side space.
[0015] The main body of the enclosure for an isolated converter according to an embodiment of the present disclosure may have a length in the vertical direction longer than a length in the horizontal direction so that an area of a bottom surface facing an installation surface is minimized.
[0016] In addition, in order to solve the above technical problems, the isolated converter according to an embodiment of the present disclosure may include an enclosure as described above; a primary-side core and a primary-side PCB disposed in the primary-side space; and a secondary-side core and a secondary-side PCB disposed in the secondary-side space.
[0017] In an embodiment of the present disclosure, the primary-side core and the secondary-side core may be disposed on the partition wall side.
[0018] In an embodiment of the present disclosure, the primary-side PCB and the secondary-side PCB may be disposed on opposite sides of the primary-side core and the secondary-side core, respectively.
[0019] In an embodiment of the present disclosure, an elastic pressing part maintaining a distance between the primary-side core and the secondary-side core may be further included.
[0020] The above disclosure provides a partition wall that divides a primary-side space and a secondary-side space, and an upper wall and a lower wall above and below the partition wall, and thus, can maximize the creepage distance compared to an enclosure of the same standard, thereby minimizing the overall size while maintaining the isolation performance.
[0021] In addition, by providing an enclosure to maximize the creepage distance compared to an enclosure of the same standard, the present disclosure can reduce the manufacturing cost because the enclosure and the isolated converter using the enclosure can be miniaturized.

Description

BRIEF DESCRIPTION OF THE DRAWINGS
[0022] FIG. **1** is a perspective view of an isolated converter according to an exemplary embodiment of the present disclosure.
[0023] FIG. **2** is a cross-sectional view of an isolated converter according to an exemplary embodiment of the present disclosure.
[0024] FIG. **3** is a cross-sectional perspective view of an enclosure for an isolated converter according to an exemplary embodiment of the present disclosure.
[0025] FIG. **4** is a perspective view of an enclosure for an isolated converter according to an exemplary embodiment of the present disclosure.
[0026] FIG. **5** is an exemplary view for explaining the creepage distance of an enclosure for an isolated converter according to an exemplary embodiment of the present disclosure.
DESCRIPTION OF SYMBOLS
[0027] **10**: isolated converter **11**: first converter [0028] **12**: second converter **100**: enclosure for an isolated converter [0029] **110**: main body **111**: primary-side space [0030] **112**: partition wall **113**: secondary-side space [0031] **114**: upper wall **114***a*, **116***a:* first stepped portion [0032] **114***b*, **116***b:* second stepped portion **114***c*, **116***c:* groove portion [0033] **115***a*, **117***a:* first fastening groove **115***b*, **117***b:* second fastening groove [0034] **115***c*, **117***c:* third fastening groove **116**: lower wall [0035] **120**: side plate **130**: input/output terminal
DETAILED DESCRIPTION
[0036] Hereinafter, in order to fully understand the configuration and effects of the present disclosure, preferred embodiments of the present disclosure will be described with reference to the

accompanying drawings. However, the present disclosure is not limited to the embodiments disclosed below, and may be embodied in various forms and various modifications may be made. Rather, the description of the present disclosure is provided so that the present disclosure will be thorough and complete and will fully convey the scope of the disclosure to those of ordinary skill in the art. In the accompanying drawings, the size of the elements is enlarged compared to actual ones for the convenience of description, and the ratio of each element may be exaggerated or reduced.

[0037] Terms such as ‘first’ and ‘second’ may be used to describe various elements, but, the above elements should not be limited by the terms above. The above terms may be used only for the purpose of distinguishing one element from another. For example, without departing from the scope of the present disclosure, a ‘first element’ may be named a ‘second element’ and similarly, a ‘second element’ may also be named a ‘first element.’ In addition, expressions in the singular include plural expressions unless explicitly expressed otherwise in the context. Unless otherwise defined, terms used in the embodiments of the present disclosure may be interpreted as meanings commonly known to those of ordinary skill in the art.

[0038] Hereinafter, an isolated converter and an enclosure for an isolated converter of an embodiment of the present disclosure will be described with reference to the drawings.

[0039] FIG. **1** is a perspective view of an isolated converter according to an exemplary embodiment of the present disclosure, and FIG. **2** is a cross-sectional view according to an exemplary embodiment of the present disclosure.

[0040] As shown in FIGS. **1** and **2**, the isolated converter **10** according to an embodiment of the present disclosure may include an enclosure **100** for an isolated converter, a primary-side core **11**, a secondary-side core **12**, a primary-side PCB **13** and a secondary-side PCB **14**.

[0041] In particular, the main body **110** of the enclosure **100** for an isolated converter has a rectangular parallelepiped shape in which the vertical length is longer than the horizontal length. This shape of the main body **110** is more advantageous for isolation.

[0042] If the horizontal length is longer than the vertical length, the area where the lower surface of the SMPS faces the fixed bottom surface is relatively widened, and the support inevitably becomes larger, which is more disadvantageous to isolation.

[0043] The enclosure **100** for an isolated converter may include a main body **110**, a side plate **120**, and an input/output terminals **130**, for accommodating the primary-side core **11**, the secondary-side core **12**, the primary-side PCB **13** and the secondary-side PCB **14**, which are the main components of the isolated converter **10**. Here, the enclosure **100** for an isolated converter is not limited to the isolated converter and may be applied to a power semiconductor transformer.

[0044] In addition, elastic pressing parts **15** and **16** for fixing the primary-side core **11** and the secondary-side core **12** to maintain magnetic force may be included.

[0045] The main body **110** may be formed in a shape divided into two parts. For example, the main body **110** may be formed in a quadrangular shape.

[0046] The main body **110** may include a primary-side space **111** for accommodating the primary-side core **11** and the primary-side PCB **13** and a secondary-side space **113** for accommodating the secondary-side core **12** and the secondary-side PCB **14**. In the drawing, a space on the left side of a partition wall **112** may be the primary-side space **111** and a space on the right side of the partition wall **112** may be the secondary-side space **113**.

[0047] As shown in FIG. **2**, a partition wall **112** may be provided inside the main body **110** to vertically separate the primary-side space **111** and the secondary-side space **113**. That is, the primary-side space **111** and the secondary-side space **113** may be divided by the partition wall **112** provided inside the main body **110**.

[0048] In addition, the main body **110** may include an upper wall **114** provided on the upper side of the partition wall **112** and a lower wall **116** provided on the lower side of the partition wall **112**.

[0049] The side plate **120** may be provided on both sides of the main body **110**. The side plate **120** may be provided to seal the primary-side space **111** and the secondary-side space **113** separated from the main body **110** by the partition wall **112**.

[0050] The input/output terminals **130** may be provided on the outside of the side plate **120**. For example, in the drawing, the left side of the main body **110** may be an input terminal, and the right side of the main body **110** may be an output terminal. That is, the input terminal may be provided on the primary side of the isolated converter **10**, and the output terminal may be provided on the secondary side of the isolated converter **10**.

[0051] The primary-side core **11** may be disposed in the primary-side space **111** of the enclosure **100** for an isolated converter. Here, the primary-side core **11** may be disposed on the partition wall **112** side within the primary-side space **111**.

[0052] The secondary-side core **12** may be disposed in the secondary-side space **113** of the enclosure **100** for an isolated converter. Here, the secondary-side core **12** may be disposed on the partition wall **112** side within the secondary-side space **113**.

[0053] As shown in FIG. **2**, the primary-side core **11** and the secondary-side core **12** may be provided to face each other with the partition wall **112** interposed therebetween.

[0054] A primary-side coil **17** and a secondary-side coil **18** are wound around the primary-side core **11** and the secondary-side core **12**, respectively, and the primary-side core **11** and the secondary-side core **12** are spaced apart for electrical isolation. The separation distance at this time is the maximum distance that can maintain a certain level of magnetic connection strength.

[0055] In this state, if the gap between the primary-side core **11** and the secondary-side core **12** is slightly widened due to impact or assembly play, the magnetic connection strength is weakened, so it is necessary to maintain the correct position.

[0056] To this end, in the present disclosure, the primary-side core **11** and the secondary-side core **12** are fixed by elastic pressing parts **15** and **16**, respectively, to fix their positions. The primary-side elastic pressing part **15** presses the primary-side core **11** in the direction of the secondary-side core **12** to prevent an increase in the distance between the primary-side core **11** and the secondary-side core **12** in advance. The same applies to the secondary-side elastic pressing part **16**. Therefore, it is possible to prevent the magnetic connection from being weakened due to external impact or assembly play. In addition, when only the elastic pressing parts **15** and **16** are removed, the primary-side core **11** and the secondary-side core **12** can be separated, which is advantageous for maintenance in the future.

[0057] The primary-side PCB **13** may be disposed in the primary-side space **111** of the enclosure **100** for an isolated converter. Here, the primary-side PCB **13** may be disposed on the opposite side of the primary-side core **11** within the primary-side space **111**. That is, the primary-side PCB **13** may be disposed on the input/output terminals **130** side. In addition, passive elements and switching elements constituting the converter circuit may be mounted to the primary-side PCB **13**.

[0058] The secondary-side PCB **14** may be disposed in the secondary-side space **113** of the enclosure **100** for an isolated converter. Here, the secondary-side PCB **14** may be disposed on the opposite side of the secondary-side core **12** within the secondary-side space **113**. That is, the secondary-side PCB **14** may be disposed on the input/output terminals **130** side. In addition, passive elements and switching elements constituting the converter circuit may be mounted to the secondary-side PCB **14**.

[0059] Hereinafter, the enclosure **100** for an isolated converter according to an embodiment of the present disclosure will be described in more detail with reference to FIGS. **3** to **5**.

[0060] FIG. **3** is a cross-sectional perspective view of an enclosure for an isolated converter according to an exemplary embodiment of the present disclosure, and FIG. **4** is a perspective view of an enclosure for an isolated converter according to an exemplary embodiment of the present disclosure.

[0061] As shown in the drawings, the partition wall **112** dividing the primary-side space **111** and the secondary-side space **113** may be provided at the center of the main body **110**. That is, the main body **110** may have a bilaterally symmetrical structure around the partition wall **112**.

[0062] In the drawings and the description below, it should be understood that the description of the primary-side space **111** can be equally applied to the secondary-side space **113**.

[0063] As shown in FIG. **3**, the primary-side space **111** and the secondary-side space **113** may be defined by an upper wall **114**, a partition wall **112**, and a lower wall **116**. That is, as shown in FIG. **4**, the enclosure **100** for an isolated converter may have a structure in which both sides are open, and may include a primary-side space **111** and a secondary-side space **113** on both sides around the partition wall **112**.

[0064] In each of the primary-side space **111** and the secondary-side space **113**, a first stepped portion **114***a* forming a step difference with the upper wall **114** may be provided. Here, the first stepped portion **114***a* may be provided with a step inward from the upper wall **114**. That is, the first stepped portion **114***a* may have a greater thickness than the upper wall **114**.

[0065] Similarly, in each of the primary-side space **111** and the secondary-side space **113**, a first stepped portion **116***a* forming a step difference with the lower wall **116** may be provided. Here, the first stepped portion **116***a* may be provided with a step inward from the lower wall **116**. That is, the first stepped portion **116***a* may have a greater thickness than the lower wall

[0066] As shown in FIG. **2**, in the primary-side space **111**, a primary-side core **11** may be disposed between the first stepped portion **114***a* of the upper wall **114** and the first stepped portion **116***a* of the lower wall **116**. Similarly, in the secondary-side space **113**, a secondary-side core **12** may be disposed between the first stepped portion **114***a* of the upper wall **114** and the first stepped portion **116***a* of the lower wall **116**.

[0067] Here, a first fastening groove **115***a* may be provided in the first stepped portion **114***a* provided on the upper wall **114**. The first fastening groove **115***a* may be provided on the opposite side of the partition wall **112** at the center of the first stepped portion **114***a*. That is, the first fastening groove **115***a* may be exposed to the outside from a groove portion **114***c*.

[0068] Similarly, a first fastening groove **117***a* may be provided in the first stepped portion **116***a* provided on the lower wall **116**. The first fastening groove **117***a* may be provided on the opposite side of the partition wall **112** at the center of the first stepped portion **116***a*. That is, the first fastening groove **117***a* may be exposed to the outside from a groove portion **116***c*.

[0069] As shown in FIG. **2**, a fastening screw (not shown) is inserted and coupled into the first fastening grooves **115***a* and **117***a* in a state where the primary-side core **11** is disposed between the first stepped portion **114***a* of the upper wall **114** and the first stepped portion **116***a* of the lower wall **116** in the primary-side space **111**, whereby the primary-side core **11** may be coupled to the partition wall **112** side within the primary-side space **111**.

[0070] Similarly, a fastening screw (not shown) is inserted and coupled into the first fastening grooves **115***a* and **117***a* in a state where the secondary-side core **12** is disposed between the first stepped portion **114***a* of the upper wall **114** and the first stepped portion **116***a* of the lower wall **116** in the secondary-side space **113**, whereby the secondary-side core **12** may be coupled to the partition wall **112** side within the secondary-side space **113**.

[0071] In each of the primary-side space **111** and the secondary-side space **113**, a second stepped portion **114***b* provided on the opposite side of the partition wall **112** from the first stepped portion **114***a* may be provided. Here, the second stepped portion **114***b* may have the same thickness as the first stepped portion **114***a*. In addition, the second stepped portion **114***b* may have a groove portion **114***c* in the center.

[0072] For example, the second stepped portion **114***b* may be provided so that the distance between the distal end and the partition wall **112** is greater than half of the primary-side space **111** or the secondary-side space **113**. That is, the distal end of the second stepped portion **114***b* may be disposed closer to the side plate **120** than the partition wall **112** in each of the primary-side space **111** and the secondary-side space **113**. Here, the distal end means a portion where a step is formed between the second stepped portion **114***b* and the upper wall **114**.

[0073] Similarly, in each of the primary-side space **111** and the secondary-side space **113**, a second stepped portion **116***b* provided on the opposite side of the partition wall **112** from the first stepped portion **116***a* may be provided. Here, the second stepped portion **116***b* may have the same thickness as the first stepped portion **116***a*. In addition, the second stepped portion **116***b* may have a groove portion **116***c* in the center.
[0074] For example, the second stepped portion **116***a* may be provided so that the distance between the distal end and the partition wall **112** is greater than half of the primary-side space **111** or the secondary-side space **113**. That is, the distal end of the second stepped portion **116***b* may be disposed closer to the side plate **120** than the partition wall **112** in each of the primary-side space **111** and the secondary-side space **113**. Here, the distal end means a portion where a step is formed between the second stepped portion **116***b* and the lower wall **116**.
[0075] As shown in FIG. **2**, in the primary-side space **111**, a primary-side PCB **13** may be disposed between the distal end of the second stepped portion **114***b* of the upper wall **114** and the distal end of the second stepped portion **116***b* of the lower wall **116**. Similarly, in the secondary-side space **113**, a secondary-side PCB **14** may be disposed between the distal end of the second stepped portion **114***b* of the upper wall **114** and the distal end of the second stepped portion **116***b* of the lower wall **116**.
[0076] Here, a second fastening groove **115***b* may be provided in the second stepped portion **114***b* provided on the upper wall **114**. The second fastening groove **115***b* may be provided on the opposite side of the partition wall **112** from both sides of the second stepped portion **114***b*. That is, the second fastening groove **115***b* may be exposed to the outside from a side of the partition wall **112**.
[0077] Similarly, a second fastening groove **117***b* may be provided in the second stepped portion **116***b* provided on the lower wall **116**. The second fastening groove **117***b* may be provided on the opposite side of the partition wall **112** from both sides of the second stepped portion **116***b*. That is, the second fastening groove **117***b* may be exposed to the outside from a side of the partition wall **112**.
[0078] As shown in FIG. **2**, a fastening screw (not shown) is inserted and coupled into the second fastening grooves **115***b* and **117***b* in a state where the primary-side PCB **13** is disposed between the distal end of the second stepped portion **114***b* of the upper wall **114** and the distal end of the second stepped portion **116***b* of the lower wall **116** in the primary-side space **111**, whereby the primary-side PCB **13** may be coupled within the primary-side space **111**.
[0079] Similarly, a fastening screw (not shown) is inserted and coupled into the second fastening grooves **115***b* and **117***b* in a state where the secondary-side PCB **14** is disposed between the distal end of the second stepped portion **114***b* of the upper wall **114** and the distal end of the second stepped portion **116***b* of the lower wall **116** in the secondary-side space **113**, whereby the secondary-side PCB **14** may be coupled within the secondary-side space **113**.
[0080] In addition, third fastening grooves **115***c* and **117***c* may be provided on both sides of the main body **110**. The third fastening grooves **115***c* and **117***c* may be provided near four corners on both sides of the main body **110**.
[0081] As shown in FIG. **3**, in a state where the side plates **120** are disposed on both sides of the main body **110**, fastening screws (not shown) are inserted and coupled into the third fastening grooves **115***c* and **117***c*, whereby the side plates **120** may be coupled to both sides of the main body **110**.
[0082] Meanwhile, the standards related to the isolation performance of the enclosure **100** for an isolated converter stipulate the insulation distance so that it can be applied during product design. Here, insulation distance is divided into clearance distance and creepage distance. The clearance distance means a distance in which the primary-side core **11** and the secondary-side core **12** directly face each other. The creepage distance is a distance between the primary-side core **11** and the secondary-side core **12** and means the shortest distance along the surface of the solid dielectric.

[0083] In the enclosure **100** for an isolated converter, since the primary-side core **11** and the secondary-side core **12** are blocked by the partition wall **112** and do not face each other, the clearance distance is not defined. The creepage distance will be described in more detail with reference to FIG. **5**.
[0084] FIG. **5** is an exemplary view for explaining the creepage distance of an enclosure for an isolated converter according to an exemplary embodiment of the present disclosure.
[0085] As shown in FIG. **3**, since the primary-side core **11** and the secondary-side core **12** disposed in the enclosure **100** for an isolated converter are disposed to face each other with respect to the partition wall **112**, the creepage distance may be defined as a distance from the partition wall **112** on the primary-side space **111** side to the partition wall **112** on the secondary-side space **113** side.
[0086] For example, the creepage distance L by the enclosure **100** for an isolated converter is a distance formed by the first stepped portion **114***a*, the second stepped portion **114***b*, the upper wall **114**, and the top surface of the main body **110** with the partition wall **112** as the center.
[0087] Similarly, the creepage distance L by the enclosure **100** for an isolated converter is a distance formed by the first stepped portion **116***a*, the second stepped portion **116***b*, the lower wall **116**, and the bottom surface of the main body **110** with the partition wall **112** as the center.
[0088] Accordingly, according to an embodiment of the present disclosure, by forming the creepage distance on both sides of the enclosure **100** for an isolated converter with the partition wall **112** as the center, it is possible to maximize the creepage distance compared to an enclosure of the same standard, thereby minimizing the overall size while maintaining the isolation performance.
[0089] In addition, according to an embodiment of the present disclosure, manufacturing costs can be reduced because the enclosure **100** for an isolated converter and the isolated converter **10** using the enclosure **100** for an isolated converter can be miniaturized.
[0090] While the present disclosure has been described in connection with what is presently considered to be practical exemplary embodiments, those skilled in the art may understand that the disclosure is not limited to the disclosed embodiments, but, on the contrary, is intended to cover various modifications and equivalent arrangements included within the spirit and scope of the appended claims. Accordingly, the true technical protection scope of the present disclosure shall be determined according to the attached claims.
[0091] The present disclosure is a technology for improving the isolation performance and reducing the size of an isolated converter using the law of nature, and has industrial applicability.

Claims

1. A enclosure for an isolated converter, comprising: a main body having a primary-side space and a secondary-side space, wherein the main body comprises: a partition wall provided to vertically separate the primary-side space and the secondary- side space from each other; and an upper wall and a lower wall provided above and below the partition wall, respectively.
2. The enclosure for an isolated converter of claim 1, wherein the partition wall is provided in the center of the main body.
3. The enclosure for an isolated converter of claim 1, further comprising side plates provided on both sides of the main body.
4. The enclosure for an isolated converter of claim 3, further comprising input/output terminals provided on the outside of the side plates.
5. The enclosure for an isolated converter of claim 1, wherein the primary-side space and the secondary-side space are defined by the upper wall, the lower wall and the partition wall.
6. The enclosure for an isolated converter of claim 1, further comprising a first stepped portion provided from the partition wall to form a step difference with the upper wall and the lower wall, in each of the primary-side space and the secondary-side space.

7. The enclosure for an isolated converter of claim 6, further comprising a second stepped portion provided from the first stepped portion to the opposite side of the partition wall and having a groove portion in the center, in each of the primary-side space and the secondary-side space.
8. The enclosure for an isolated converter of claim 1, wherein the main body has a length in the vertical direction longer than a length in the horizontal direction so that an area of a bottom surface facing an installation surface is minimized.
9. An isolated converter, comprising: an enclosure according to any one of claims **1** to **8**; a primary-side core and a primary-side PCB disposed in the primary-side space; and a secondary-side core and a secondary-side PCB disposed in the secondary-side space.
10. The isolated converter of claim 9, wherein the primary-side core and the secondary-side core are disposed on the partition wall side.
11. The isolated converter of claim 9, wherein the primary-side PCB and the secondary-side PCB are disposed on opposite sides of the primary-side core and the secondary-side core, respectively.
12. The isolated converter of claim 10, further comprising an elastic pressing part maintaining a distance between the primary-side core and the secondary-side core.